

EXACT HISTORY DYNAMICS, DOMAIN-SAFE MEMORY, AND UNIVERSAL EXCESS PATH PRESSURE

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ABSTRACT. The prepared Sinai future is generated by one unchanged past kernel. A small-distance coupling, a Lyapunov drift, and a common small set produce a patched weak-Harris metric and a weighted spectral gap. A fixed multiplier Banach algebra gives an analytic Feynman–Kac and eigenfunction-Doob tower. The unsmoothed renewal resolvent is continued with integrable vertical bounds, and a common-domain Schur complement yields the complete forced Mori–Zwanzig equation, including the unresolved initial component. The martingale rough-path tangent is proved in the same history topology.

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1. ROUND-SEVENTEEN POSITIVE CLOSURE: ONE GLOBAL HISTORY KERNEL AND FORCED COMPRESSED MEMORY

The history dynamics is constructed globally from the legal return graph. No partition of unity is used to mix local kernels. Weak-Harris contraction is proved in a patched distance from a derived drift, a derived small-set minorization, and the summable-variation coupling. The memory identity is

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a common-domain block-resolvent theorem and retains the unresolved initial forcing.

1.1. The global past kernel. Let Ω^- be the space of legal left-infinite return-edge histories with the current suspension residual. Put

$$d_\beta(x, y) = \sum_{j \geq 0} e^{-\beta j} (1 \wedge d_E(x_{-j}, y_{-j})), \quad W(x) = \exp\{\eta(r(x_0) + \tau(x_0) + \mathfrak{s}(x_0))\}.$$

The reference Gibbs conditional probability of the next edge is denoted $g(e \mid x)$. Summable variations give

$$\sum_{n \geq 1} \text{var}_n(\log g) < \infty.$$

Appending a legal edge and shifting the past defines one global Markov kernel P on Ω^- .

Lemma 1.1 (Derived one-step coupling). *There are $C, c > 0$ such that, if two histories agree through depth N , then their next-edge kernels have total variation distance at most Ce^{-cN} . Under a maximal coupling, a successful match multiplies d_β by $e^{-\beta}$; a failed match has probability at most $Cd_\beta(x, y)^\alpha$ for some $\alpha > 0$.*

Proof. The Gibbs ratio of two next-edge probabilities is the exponential of the variation of the induced potential on histories sharing their last N edges. Summable variation bounds the logarithmic ratio by Ce^{-cN} , and Pinsker's elementary two-measure estimate gives the total variation bound. Choose N comparable with $|\log d_\beta|$. If the next edges match, all old discrepancies shift one coordinate into the past, giving the factor $e^{-\beta}$. $\square$

Lemma 1.2 (Lyapunov drift and a common small set). *For sufficiently small $\eta > 0$,*

$$PW \leq \lambda W + C, \quad \lambda < 1.$$

For every $R < \infty$, the set $\{W \leq R\}$ is n_R -small: there are a probability ν_R and $\epsilon_R > 0$ such that

$$P^{n_R}(x, \cdot) \geq \epsilon_R \nu_R(\cdot) \quad \text{for all } W(x) \leq R.$$

Proof. The exponential return/roof/singularity moment of the induced Gibbs kernel implies a Foster–Lyapunov estimate after reducing η ; the current edge is shifted out and the new exponential moment is uniformly bounded. On a Lyapunov sublevel only finitely many edge cylinders carry all but a fixed small probability. The finite connector property supplies a common legal word from each such cylinder to one fixed magnet cylinder. Bounded Gibbs distortion gives a uniform positive lower bound on the probability of that word and on a common terminal subcylinder, proving minorization. $\square$

Define $d_\epsilon = 1 \wedge \epsilon^{-1} d_\beta^\alpha$ and

$$\tilde{d}(x, y) = \sqrt{d_\epsilon(x, y)(1 + \delta W(x) + \delta W(y))}.$$

Theorem 1.3 (Global weak-Harris spectral theorem). *For suitable ϵ, δ there are n_0 and $\rho < 1$ such that*

$$\mathcal{W}_{\tilde{d}}(P^{n_0}(x, \cdot), P^{n_0}(y, \cdot)) \leq \rho \tilde{d}(x, y).$$

Consequently P has a unique invariant probability ν , a spectral gap on the weighted Lipschitz space

$$\mathcal{B} = \{f : \|f\|_W + \text{Lip}_{\tilde{d}} f < \infty\},$$

and exponentially summable correlations. This kernel is the original global Doob/history kernel; no chart mixture has been introduced.

Proof. For small d_ϵ , use the coupling of Lemma 1.1; the failed-match term is absorbed by choosing ϵ small. For large distance but bounded W , use the common small-set coupling from Lemma 1.2. Outside that set, the Lyapunov drift contracts the weight factor. The three estimates are exactly the weak-Harris contraction in the patched distance. Iteration gives the spectral gap and correlation bound on $\mathcal{B}$. $\square$

1.2. One fixed multiplier algebra and Doob tower. Fix $0 < \eta_0 < \eta$ and define

$$\mathcal{A} = \left\{ V : \|V\|_{\mathcal{A}} := \sup_x \frac{|V(x)|}{1 + \log W(x)} + \text{Lip}_{\tilde{d}} V < \infty \right\}.$$

Let $\mathcal{B}_\theta$ be the same weighted Lipschitz space with weight W^θ , $0 \leq \theta \leq 1$.

Theorem 1.4 (Fixed analytic Feynman–Kac chart). *There is $r_0 > 0$ and fixed $0 < \theta < \theta' < 1$ such that*

$$P_V f = P(e^V f)$$

is an analytic family in $\|V\|_{\mathcal{A}} < r_0$ from $\mathcal{B}_\theta$ to itself and from $\mathcal{B}_\theta$ to $\mathcal{B}_{\theta'}$. It has a simple positive eigenvalue $e^{\Lambda(V)}$, a positive eigenfunction h_V , and a dual eigenmeasure ν_V . The normalized kernel

$$\hat{P}_V f = e^{-\Lambda(V)} h_V^{-1} P_V(h_V f)$$

is Markov and satisfies the exact nonlinear tower

$$\mathcal{E}_{m+n}^V F = \mathcal{E}_m^V(\mathcal{E}_n^V F).$$

Proof. The inequality $|V| \leq r_0(1 + \log W)$ implies $e^{|V|} W^\theta \leq C W^{\theta'}$ when $r_0 < \theta' - \theta$. The Lipschitz product estimate and Theorem 1.3 give a uniform bounded multiplier map on the fixed pair of spaces. The exponential power series converges in operator norm, hence is analytic. Analytic perturbation of the isolated positive eigenvalue gives Λ, h_V, ν_V . The displayed Doob transform is Markov by the eigenfunction equation, and its ordinary Markov tower becomes the nonlinear tower after logarithmic conjugation. $\square$

1.3. Martingale and rough-path limit. For centered $g \in \mathcal{B}$, the Poisson series $h = \sum_{n \geq 0} P^n g$ converges in $\mathcal{B}$, and

$$d_{n+1} = h(X_{n+1}) - Ph(X_n)$$

is a martingale difference.

Theorem 1.5 (Enhanced invariance principle). *For every finite vector of centered observables in $\mathcal{B}$ with a slightly higher W -moment, the suspension additive functional and its iterated integral converge jointly to a Brownian rough path. The covariance is the absolutely convergent Green–Kubo series and the antisymmetric area drift is the convergent martingale–coboundary area series. The conclusion is stable under conditioning on a compact past cylinder.*

Proof. The spectral gap gives the Poisson solution and an $L^{2+\epsilon}$ martingale–coboundary decomposition. The martingale functional CLT and its iterated-integral version apply; the coboundary contributes only the explicit area correction. Exponential roof moments control suspension inversion and the final residual roof. Conditioning on a compact past cylinder changes the initial law by a uniformly bounded density, so the same Lindeberg and bracket estimates hold. $\square$

1.4. Raw renewal resolvent. Let $R_V(z)$ be the first-return operator weighted by $e^{-z\tau+V}$. Its matrix coefficients are the Laplace–Fourier transforms controlled by A2.

Theorem 1.6 (Unsmoothed suspension resolvent). *For V in a compact subball of $\mathcal{A}$, the renewal identity*

$$(z - L_V)^{-1} = A_V(z) + B_V(z)(I - R_V(z))^{-1}C_V(z)$$

holds from a common graph domain D to $L^2(\nu_V)$. The terms are meromorphic in $\Re z > -\gamma$, and after the finitely many resonances are removed their matrix coefficients and first two source derivatives are integrable on vertical lines. The bounds are uniform in V .

Proof. Decompose a suspension path into its initial residual, complete returns, and final residual. Fubini on the common exponential roof weight gives the renewal identity first for $\Re z$ large. The A2 trace-free matrix-coefficient Fourier estimate gives an integrable bound for the complete-return series; the residual operators are bounded by the exponential roof moment. Analytic Fredholm reduction on the isolated finite-dimensional leading spectral subspace gives the meromorphic continuation. The same estimates hold after two source derivatives because the multiplier algebra is fixed. $\square$

1.5. Common-domain forced compressed memory. Let $H_V = L^2(\nu_V)$, let L_V be the generator of the normalized suspension semigroup, and let P_V be the orthogonal projection onto a finite-dimensional resolved space

$\mathcal{R}_V \subset D(L_V^2)$. The spaces are transported to one reference Hilbert space by the square-root density isometry. Put $Q_V = I - P_V$ and

$$C_V(z) = P_V(z - L_V)^{-1}P_V.$$

Lemma 1.7 (Right-half-plane compression bound). *For $\Re z \geq \sigma > 0$, $C_V(z)$ is invertible on $\mathcal{R}_V$. On compact vertical contour segments its least singular value is uniformly positive, while for large $|z|$,*

$$C_V(z) = z^{-1}P_V + O(|z|^{-2}).$$

Proof. The normalized Markov semigroup is an $L^2(\nu_V)$ -contraction, so L_V is dissipative. If $x \in \mathcal{R}_V$ and $y = (z - L_V)^{-1}x$, then

$$\Re \langle C_V(z)x, x \rangle = \Re \langle y, (z - L_V)y \rangle \geq \sigma \|y\|^2.$$

Thus $C_V(z)x = 0$ implies $x = 0$; finite dimensionality gives invertibility. Compactness of the source set and contour gives the uniform minimum singular value. The large- z expansion follows from $\mathcal{R}_V \subset D(L_V^2)$. $\square$

Define

$$\widehat{K}_V(z) = zP_V - P_V L_V P_V - C_V(z)^{-1}.$$

Theorem 1.8 (Forced Mori–Zwanzig identity). *For $A \in D(L_V)$, the resolved trajectory $x_A(t) = P_V e^{tL_V} A$ satisfies, in $\mathcal{R}_V$,*

$$\dot{x}_A(t) = P_V L_V P_V x_A(t) + \int_0^t K_V(t-s)x_A(s) ds + \eta_A(t),$$

where

$$\widehat{\eta}_A(z) = C_V(z)^{-1}P_V(z - L_V)^{-1}Q_V A.$$

The kernel is the causal inverse Laplace transform of $\widehat{K}_V$. After adjoining the finitely many left-half-plane transmission-zero residue modes, the residual kernel and forcing are exponentially integrable. If $A \in \mathcal{R}_V$, the forcing vanishes.

Proof. The Laplace transform of x_A is

$$X_A(z) = C_V(z)P_V A + P_V(z - L_V)^{-1}Q_V A.$$

Multiply by $C_V(z)^{-1}$ and rearrange to obtain the transform of the stated Volterra equation, including its initial value and forcing. All products are defined because $\mathcal{R}_V \subset D(L_V^2)$. Lemma 1.7 gives the inverse on the right half-plane. Theorem 1.6 gives meromorphic continuation. A zero of the continued compression is promoted, with its full Riesz residue range, to the resolved space; the Schur complement then has no such pole. Finitely many steps remove all zeros in a fixed strip, and contour shifting gives exponential integrability. $\square$

Theorem 1.9 (Round-seventeen A4 closure). *The prepared Sinai history has one global weak-Harris kernel, one fixed analytic Feynman–Kac algebra, a conditional enhanced invariance principle, an unsmoothed renewal resolvent, and a common-domain forced compressed-memory equation with quantitative inverse bounds.*

Proof. Combine Theorems 1.3, 1.4, 1.5, 1.6, and 1.8. □

SCOPE AND VERIFICATION STATUS

This manuscript states a positive theorem reconstruction in the model-specific regular regime declared in its controlling theorems. Cross-paper inputs follow the Round-Seventeen dependency ledger. Repository source identity, proof-structure, counterexample, and build checks are internal reproducibility gates and do not replace independent external mathematical review.

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